

Geological context changes the validation question

Coverage status describes the current coding resolution. It is not a prevalence estimate or a score.

deep saline aquifers

Moderate coverage

Saline-aquifer labels do not by themselves encode basin, depth, heterogeneity, or brine-chemistry diversity.

depleted oil and gas reservoirs

Not reliably stratified

Reservoir depletion state and legacy-well conditions cannot be reconstructed consistently from current coding.

offshore and regional screening

Thin coverage

Regional screening and offshore storage formation evidence are not equivalent.

faulted reservoirs

Thin coverage

One synthetic connectivity study cannot represent fault transmissibility, slip risk, and site diversity.

EOR-storage hybrids

Not reliably stratified

Storage containment and oil-recovery objectives can impose different target variables and evidence standards.

field projects and controlled experimental benchmarks

Not reliably stratified

Field anchors, field-inspired models, and ML evaluated on field or controlled data must remain distinct.

NR means that the context cannot be reconstructed consistently from the locked fields; it does not mean zero evidence.